<u>Creature Customizer Class Methods:</u>

-all methods are get/set/remove

<u>Creature Class Methods:</u>

1. METHOD addDebuff(debuff: Debuff): void

```
    IF this.creature.resistances DOES NOT CONTAIN debuff THEN
        this.creature.debuffs.add(debuff)
    ELSE
        display blocked debuff message
    END IF
END METHOD
```

2. METHOD removeDebuff(debuff: Debuff): void

```
    this.creature.animations.happyAnimation.play()
    this.creature.debuffs.remove(debuff)
END METHOD
```

3. METHOD attack(): Integer

```
    selectedEnemy.health -= creature.damage
    this.creature.animations.attackAnimation.play()

    totalDamageDealt = this.creature.baseDamage

    selectedEnemy.armor -= totalDamageDealt

    healthDamageDealt = 0
    IF selectedEnemy.armor < 0 THEN
        healthDamageDealt = ABS(selectedEnemy.armor)
        selectedEnemy.armor = 0
    END IF

    IF healthDamageDealt > 0 THEN
        selectedEnemy.health -= healthDamageDealt
    END IF


    IF selectedEnemy.health <= 0 THEN
        selectedEnemy.removeFromTurnStack()
        selectedEnemy.animations.deathAnimation.play()
    END IF
```

```
    RETURN totalDamageDealt
END METHOD
```

Player Class Methods:

1. METHOD addDebuff(debuff: Debuff): void

```
    IF this.player.resistances DOES NOT CONTAIN debuff THEN
        player.animations.sadAnimation.play()
        this.player.debuffs.add(debuff)
    ELSE
        display blocked debuff message
    END IF
END METHOD
```

2. METHOD removeDebuff(debuff: Debuff): void

```
    this.player.animations.happyAnimation.play()
    this.player.debuffs.remove(debuff)
END METHOD
```

3. METHOD useItem(item: Item, friendlyEntity: Entity)

```
    IF item.itemType == HEALING THEN
        targetEntity.heal(item.healValue)
        removeItem(item)
    END IF
END METHOD
```

4. METHOD heal(healValue: Integer): void

```
    this.player.health += healValue
    IF this.player.health > this.player.maxHealth
        This.player.health = this.player.maxHealth
    END IF
END METHOD
```

5. METHOD attack(): Integer

```
    player.animations.attackAnimation.play()

    totalDamageDealt = this.player.baseDamage
```

```
    selectedEnemy.armor -= totalDamageDealt

    healthDamageDealt = 0
    IF selectedEnemy.armor < 0 THEN
        healthDamageDealt = ABS(selectedEnemy.armor)
        selectedEnemy.armor = 0
    END IF

    IF healthDamageDealt > 0 THEN
        selectedEnemy.health -= healthDamageDealt
    END IF


    IF selectedEnemy.health <= 0 THEN
        selectedEnemy.removeFromTurnStack()
        selectedEnemy.animations.deathAnimation.play()
    END IF
    RETURN totalDamageDealt
END METHOD
```

EnemyEntity Class Methods:

1. METHOD addDebuff(debuff: Debuff): void

```
    this.enemyEntity.animations.angryAnimation.play()
    this.enemyEntity.debuffs.add(debuff)
END METHOD
```

2. METHOD removeDebuff(debuff: Debuff): void

```
    this.enemyEntity.animations.tauntAnimation.play()
    this.enemyEntity.debuffs.remove(debuff)
END METHOD
```

3. METHOD attack(): Integer

```
    target = findOptimalTarget()
    totalDamageDealt = this.enemyEntity.baseDamage

    target.armor -= totalDamageDealt

    healthDamageDealt = 0
    IF target.armor < 0 THEN
        healthDamageDealt = ABS(target.armor)
        target.armor = 0
```

```
    END IF

    IF healthDamageDealt > 0 THEN
        target.health -= healthDamageDealt
    END IF


    IF target.health <= 0 THEN
        target.removeFromTurnStack()
        target.animations.deathAnimation.play()
    END IF

    IF target.isPlayer() THEN
        endCombat()
    END IF

    RETURN totalDamageDealt

END METHOD
```

HatcheryHandler Class Methods:

1. METHOD feed(selectedFood: Food) Integer

```
    this.creature.hunger -= selectedFood.hungerRestoreValue
    removeItem(selectedFood)
    RETURN selectedFood.hungerRestoreValue
END METHOD
```

2. METHOD heal(item: Item): Integer

```
    this.creature.health += item.healValue
    IF this.creature.health > this.creature.maxHealth
        This.creature.health = this.creature.maxHealth
    END IF
    RETURN item.healValue
END METHOD
```

3. METHOD hatchEgg(selectedEgg: Egg): Creature

```
    Creature creature = selectedEgg.getRandomCreature
    addCreature(creature)
    RETURN creature
END METHOD
```

Rest of methods in hatchery handler are basic CRUD

Combat Handler Class Methods:

1. METHOD passTurn(): void
```
    turnStack[0].endTurn()
    currentTurnOwner = turnStack.pop()
    currentTurnOwner.startTurn()
END METHOD
```

2. METHOD giveDebuff(target: Entity, debuff: Debuff): void
```
    target.addDebuff(debuff)
END METHOD
```

-Rest of the methods are basic updates

Scene Manager Class Methods:

1. METHOD enterCombat(): void
```
    IF selectedCreatures.length > 0 AND verifyCreatureSelection(selectedCreatures) == true
THEN
        FOR scene in openScenes
            scene.close()
        END FOR

        combatScene.init()
    END IF
END METHOD
```

2. METHOD verifyCreatureSelection(selectedCreatures: List<Creature>): Boolean
```
    FOR creature in selectedCreatures
        IF creature.health <= 0
            RETURN false
        END IF

        IF creature.isSick
            RETURN false
        END IF

        IF creature.isExhausted
```

```
                RETURN false
            END IF

    END FOR

    RETURN true
END METHOD
```

3. METHOD getCombatReward(): CombatReward

```
    CombatReward reward = new CombatReward(level)
    IF NOT level.alreadyCompleted THEN
        RETURN reward;
    END IF

    RETURN null
END METHOD
```

-rest of methods in scene manager are basic CRUD

Combat Reward class methods:

1. METHOD getRewards(): Map<Item, Integer>

```
    Map<Item, Integer> itemToCount = generateRewardsFromLevel(currentLevelNumber)
    RETURN itemToCount
END METHOD
```

- All other methods are basic CRUD

EquipmentManager class methods:

- all methods are basic CRUD